

Enrollment No:
74, 75, 76, 77

Smart Road

Group No : 12



**BHAGWAN MAHAVIR UNIVERSITY
BHAGWAN MAHAVIR COLLEGE OF COMPUTER
APPLICATION, SURAT
M.Sc. IT (Integrated)
TY B.Sc. (Information Technology)
Semester V
Subject: 1020406503 Internet of Things**

Group: 12

[Smart Road]



Team :-

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1. Abstract :-

Smart roads built on IOT and information and communications technology can make it possible for cities and transportation authorities to collect analyze data to improve day-to-day traffic management.

Smart Road with IOT based traffic management, So IOT can help the traffic system to be automated and monitor the traffic status using the internet. Automatic street light system is smart and provides a safe night time environment for all road users including pedestrians.

Smart waste management system : In this system, a 24*7 monitoring system is designed for monitoring dumpsters.

2. Introduction:-

Roads are the arteries through which the economy pulses. By linking producers to markets, workers to jobs, students to school, and the sick to hospitals, roads are vital to any development agenda. Since 2002, the World Bank has constructed or rehabilitated more than 260,000 km of roads. It lends more for roads than for education, health, and social services combined. However, while roads bring economic and social benefits, they can also come with social costs such as pollution or deforestation. Roads are the main source which connects all the villages, cities, states and countries and enhances communication among all parts of the world.

On an average, 2 billion people use roads for their transport everyday. The present roads which we are using are the age old ones and have not been developed in any aspect. Now a days, there is up gradation in all aspects of life but roads are the only ones which have not been modernised in our country. Roads have been technologically developed in all other parts of the world based on their requirements. Infrastructure has always mattered. The industrial revolution was not just the story of cotton mills and iron foundries. It was about canals, railways and roads. Big changes to the way people and goods could move around the country boosted growth and transformed the economy. The roads which are now in practice in all parts of the world are the ones which not only provide transport but also support us by

providing power by using wind energy and solar energy. These roads are called SMART ROADS.

Smart Roads are expected to be used for four aspects:

- Innovation (facilitation of autonomous vehicles operations);
- User service quality (traffic monitoring, communication, safety);
- Infrastructure monitoring (damages detection);
- Energy source (solar panels, piezoelectric materials).

3. Project Profile:-

3.1 Goals of Project:-

Our Goal is to make Smart Road with used of these advanced Technology and also Smart Traffic signals, smart Waste Mangement System for Reduce man power and Clean Environment.

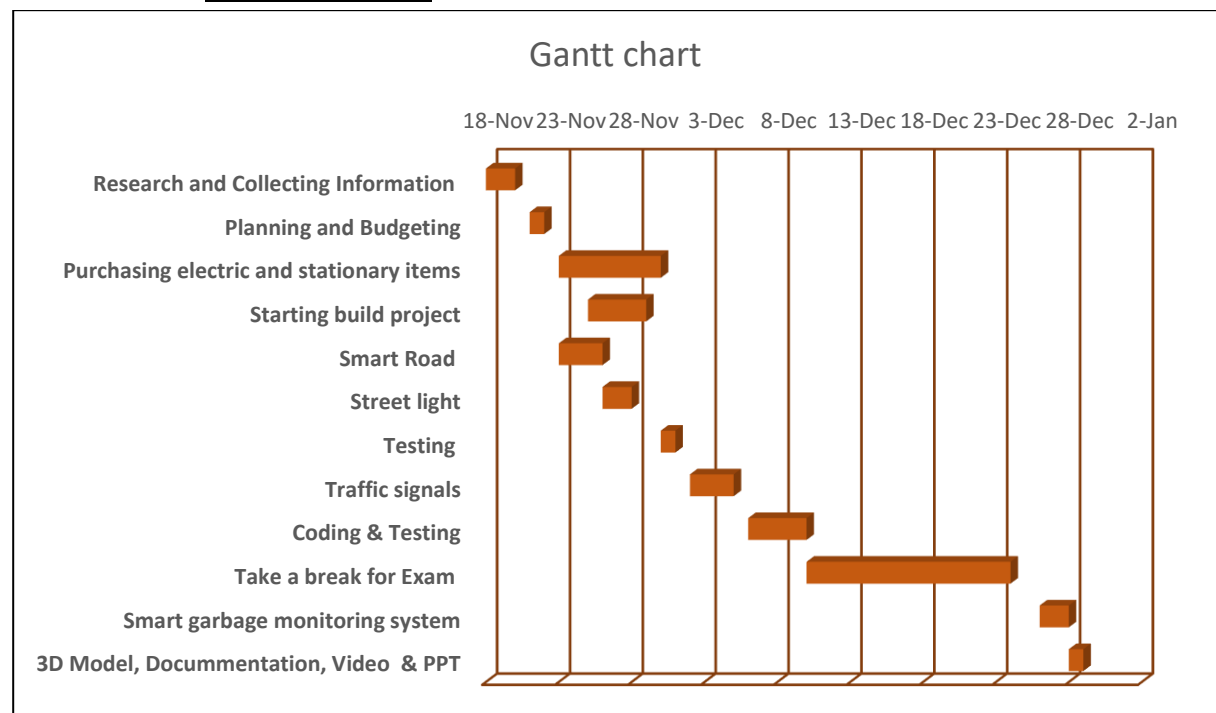
It aims to create a better smart road that offers a variety of smart features including smart traffic signals, smart street lights, smart garbage monitoring system and smart alert button. City becomes smarter thanks to these gadgets.

From this things, electricity can be save with the help of smart street lights. Secondly, environments or quality time also can be save with the help of smart traffic signals. Moreover, health and other social issues can be save or reduce with help of garbage monitoring system and smart alert button.

3.2 Project Team:-

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3.3 Gantt Chart:-



4. System Analysis:-

4.1 Requirment Analysis:-

This project is mostly required in mega cities because the number of vehicles and population increasing day to day that create a traffic problem, air or noise pollution, and other social and health issues as well.

❖ Used Items :-

1. Ultra sonic sensor
2. Wires
3. Male to female, Male to male, Female to female
4. LED lights
5. Battery
6. Breadboard
7. Arduino UNO
8. Servo motor & Motor
9. Registers
10. Piezo

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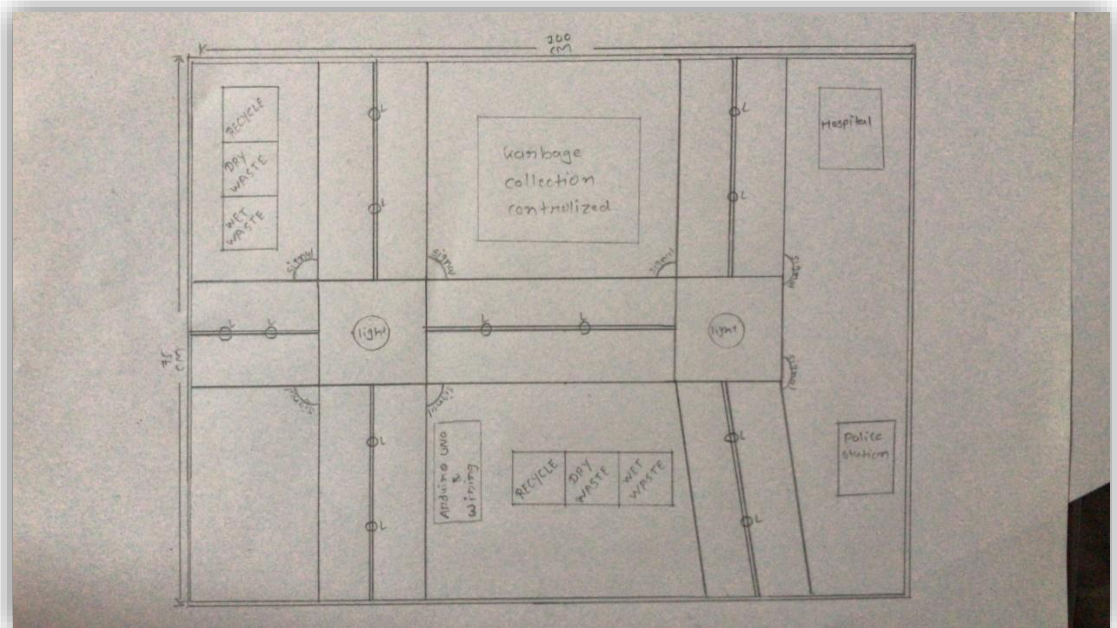
11. Buttons & Switches

12. Stationary items : Cardboard sheet, Thermocol sheet, Straws, Glue,
Matchbox, Colour etc...

5. System Design:-

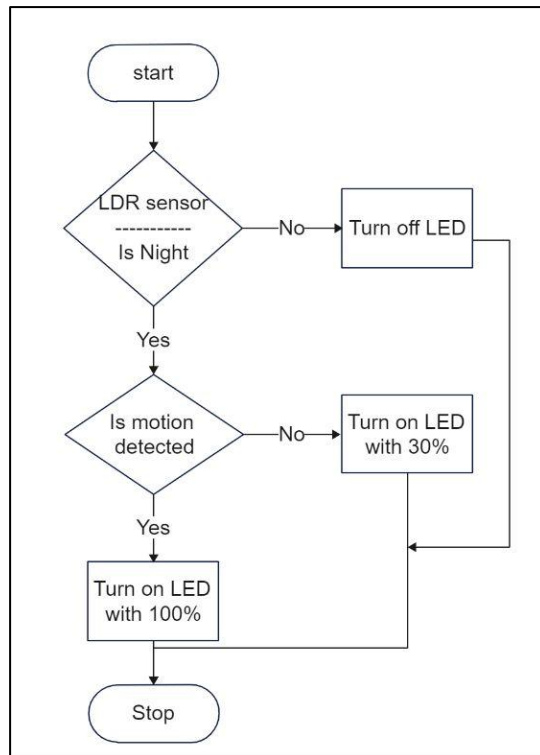
5.1 Architecture Diagram:-

This Design is to measure road Architecture

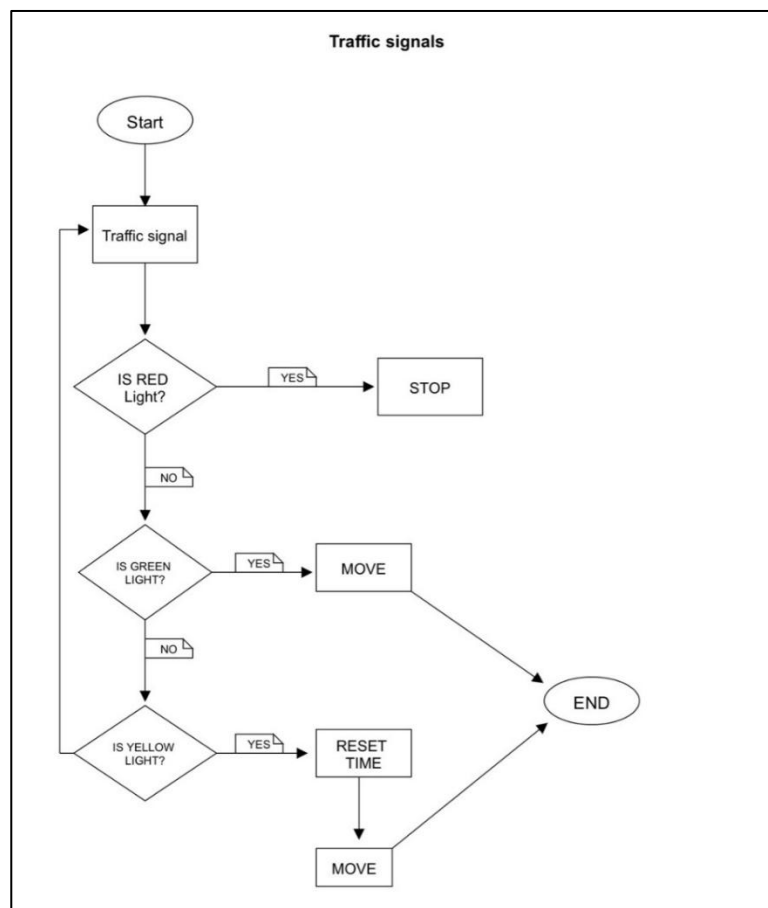


5.2 Flow Chart:-

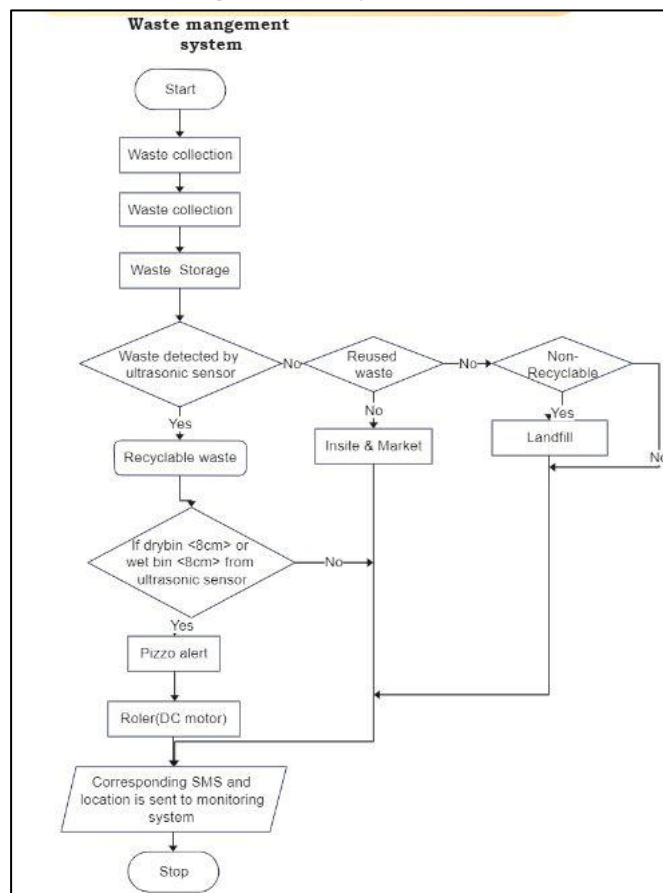
Street Light :-



Traffic Signal Flowchart :-



Waste Management System:-



5.3 Blue Print:-

❖ 3D Design (Top View):-

This model given information about project. Every parts of project has been given in this Model. The Model is very simple and informative. So, anyone can understand easily.

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6. Implementation:-

- To increase the road usability.
- Improvement of lighting in road which saves a lot of electricity and money.
- Use of digital media to transfer information.
- Follow Smart Traffic Light signals.
- Use Smart Waste Management System for Pollution Control or Clean Roads.
- Use All Smart Road Technologies.

7. Benefits & Challenges:-

❖ Benefits :-

Interactive lights:

One of the most important features of the road, baring in mind the energy challenges from today, are the interactive streetlights that switch on when cars pass by and fade out when they pass by. By using such lights millions of watts of electricity can be saved. These lights are so helpful that they are 10 times brighter than the lights which we are using now. This will help the driver see clearly and the number of accidents can be reduced. They are also cost efficient. In India, such lights would help reduce the electricity required for roads and this can be used in many other aspects.

 **Huge reduction of Energy and maintenance cost**

 **Increased public Safety from improved lighting**

 **Traffic Light timing in real-time:** The smart traffic management system helps traffic light to operate in real-time conditions. Traffic operates based on traffic congestion automatically.

 **Reduction in pollution:** Due to the good engagement of traffic flow, pollution can be reduced. People can save fuels and due to fuel saving, it leads to a reduction in air pollution.

 **Smart Waste Management System:** This method is a very useful method for stop pollution in mega cities. It can be useful for reduce man effort, pollution, fueilling or many more. Waste disposal truck driver wants to visit all the bins and check that which bins are full or

and which are empty. The garbage monitoring system that contains these all the facilities are automatically.

❖ **Difficulties faced by dwellers :-**

- ✚ **Lack of awareness:** Mannerless people throwing trash anywhere, trash can inviting so many diseases.
- ✚ **Transportation and Traffic:** The number of mobility is increasing day to day in mega cities. That create a traffic on roads, noise pollution, air pollution, time consuming.
- ✚ **Other Social and Health issues:** In cities, mobility lead a accidents, others are large violent crime, mob fighting...etc.
- ✚ Waste collection methods should be ecofriendly and should be in such a way that those methods should not effect environment.
- ✚ Waste collection methods should utilize the raw materials very efficiently and resource management should be efficient.
- ✚ **Risk in Road Construction:** Inefficient design drawings, Misinterpretation of plans, Site accidents, hazards & damages
- ✚ **Cost Overrun and Failed Estimates**
- ✚ Increased accident rates due to the non-maintenance of damaged parts of the road.

8. Future Enhancement:-

- ✚ **What smart road technologies are shaping the future of transportation?**



Proposal future works scheme.

Roadways are constantly in flux, either through repairs and construction or continual improvements. With recent developments in smart technologies, companies and research groups alike have been hard at work devising ways to make the roads safer and more driver-friendly, and enable the use of developments coming from the automotive sector. This report discusses a wide variety of these technologies and how they will continue to be developed in the near future.

+ Our role:-

we should doing something better and make a good management system with smart technology for roads.

9. Bibliography :-

For successfully completing my project, I have taken help from following websites or links:

+ www.google.com

+ www.wikipedia.com

+ Youtube

+ <https://create.arduino.cc/projecthub/sagnik2017ghosh/iot-based-smart-street-light-system-8e9929>

+ [https://www.researchgate.net/publication/360620242 ACIDENT DETECTION AND ALERT SYSTEM](https://www.researchgate.net/publication/360620242_ACIDENT_DETECTION_AND_ALERT_SYSTEM)

+ <https://www.pantechsolutions.net/iot-based-smart-waste-management-system-for-smart-city>

+ What we learnt from this project ?

There are so many problems that we are appearance in some cities such as pollution, traffic problems, safety, other social and health issues, mob crowding on roads, some mannerless activities.. or many more. So learnt that, we can do something for this type of city or roads with the help of IOT(“Internet Of Things”).